Roya Akrami

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EDUCATION

Colorado School of Mines **Graduating May 2027**
Bachelor of Science, Electrical Engineering

- Applied embedded systems, signal analysis, and hardware-software integration using Verilog, C, MPLAB X IDE, FPGA development boards, Arduino, Raspberry Pi, and custom PCBs.

NASA Lunabotics Team - Capstone Design - Electrical Subsystem Lead

- Developing electrical wiring diagrams, evaluating power-distribution architectures, and selecting controller and power-system components for an autonomous lunar excavation rover.
- Defining electrical subsystem interfaces and assessing hardware integration requirements to support reliable system operation.
- Selected from a cohort of 645 students to lead electrical subsystem development within an 11-member NASA Lunabotics capstone team.

PROFESSIONAL EXPERIENCE

Research Collaborator **February 2026 - Present**
NASA Jet Propulsion Laboratory

- Developed and trained a 1D convolutional neural network to automate anomaly detection in noisy exoplanet transit data.
- Tested and reported on software for identifying and categorizing detection errors within the Excalibur data-processing pipeline.

Research Assistant **May 2026 - August 2026**
ETH Zurich - Laboratory for Movement Biomechanics

- Developed a MATLAB-based data-processing and modeling pipeline to estimate foot placement from wearable IMU measurements.
- Evaluated model performance using statistical analysis and experimental data to support the development of a wearable motion-monitoring system.

Research Assistant **March 2025 - February 2026**
NASA Goddard Space Flight Center

- Processed and analyzed satellite emission data using Python, JupyterHub, and MATLAB to support the development of predictive models.
- Applied a 28-layer residual neural network to analyze H-Lyα intensity measurements and predict future emission patterns.
- Presented project, “A Machine Learning-based Model of the Terrestrial Exospheric Emission Using Lyman-Alpha Radiance Data Acquired by NASA's TWINS Mission,” at the American Geophysical Union 2025 conference, communicating NASA internship research in space-based machine learning.

Research Assistant **Oct 2023 - Oct 2025**
Phal Lab, Colorado School of Mines

- Designed and implemented a 2D LiDAR-based mapping system using a LattePanda computer, Ubuntu, and ROS RViz for real-time visualization and spatial mapping.
- Developed technical figures and documentation for research involving optical spectroscopy and laser-based measurement systems, resulting in becoming a coauthor for “*Rapid Vibrational Circular Dichroism Spectroscopy via Synchronized Photoelastic Modulator–Quantum Cascade Laser Integration*.”

Undergraduate Research Ambassador **Aug 2024 - Present**

Colorado School of Mines

- Selected to represent Colorado School of Mines as a student ambassador for undergraduate research in Washington, D.C.; funded by the STR Program to meet with U.S. senators and discuss research funding and policy.
- Presented to students on the value of undergraduate research and provided mentorship through email outreach, and authored articles highlighting student research initiatives.

Society of Women in Research **Feb 2025 - Present**

Founder and President

- Founded and led the only organization at Colorado School of Mines dedicated to closing the gender gap in research participation.
- Organized weekly meetings, cold-email workshops, and scholarship information sessions to connect female students with research opportunities. Presented organization to more than 100 students and parents.

PROJECTS

Optical-Link Fault Detector **August 2026**

- Developed a Python anomaly detector to flag potential optical-link degradation using simulated optical power, temperature, and bit-error measurements.

Rack and Power Capacity Planner **August 2026**

- Built a Python optimization tool to allocate network equipment across racks under space, power, and cooling constraints and estimate additional infrastructure requirements.

LANGUAGES

C2-level French • C++ • Python • MATLAB • C • JupyterHub • MPLAB x IDE • Inkscape • Excel • LaTeX • Verilog

HONORS & AWARDS

ThinkSwiss Scholarship **February 2026**

- Awarded $7,000 to conduct research overseas at ETH Zurich of Switzerland.

Travel Grant from Colorado School of Mines **December 2025**

- Awarded $400 for travel expenses to attend American Geophysical Union conference to present exospheric machine learning research conducted at the NASA Goddard Space Flight Center.

NASA and Universities Space Research Association **Spring 2025**

- Awarded a ~$10,000 undergraduate research stipend to conduct NASA-funded research.

Students Transforming Through Research Conference **Spring 2025**

- Funded to attend a research policymaking conference in Washington D.C to represent Mines as an Undergraduate Research Ambassador.

Summer Undergraduate Research Fellowship **June - Aug 2024**

- $4000 stipend to conduct faculty-mentored summer research, granted by Colorado School of Mines.

3rd Place at Spring Undergraduate Research Symposium **April 2024**

- Presented 12-minute presentation as a first-year student of vibrational circular dichroism designs at Mines Research Symposium, won 3rd place out of all presenters.

First-Year Innovation & Research Scholar Training **Aug 2023 - May 2024**

- $1000 stipend to work in a research lab, granted by Colorado School of Mines.

The Provost Award **Aug 2023 - May 2027**

- $10,000-per-year scholarship granted by Colorado School of Mines